

Homework 2: Exploring One Sample Inference

Submission Notes: Please write your answers to each question under that question (not altogether at the end). It will be easier for me to give full credit if your answer is commented well, including comments about where you are completing each sub-part of the question. For an example, see Homework 1.

1. Complete the following to explore differences in tests that use discrete (versus continuous) sampling distributions. In both parts (a) and (b), assume we aim to conduct the following test:

$$H_0 : p = 0.50$$

$$H_a : p < 0.50$$

- (a) A Binomial Test under the condition that we plan to obtain $n = 15$ observations.
 - i. Plot a Binomial($n = 15$, $p = 0.50$) distribution.

```
library(ggplot2)
library(tidyverse)

n = 15 #our sample size

p = 0.5 #our hypothesis probability

#Create tibble of associated probabilities of each observation under the binomial distribution
ggdat <- tibble(x = 1:n,
                p = dbinom(1:n,size = n, p = p))

ggplot(data = ggdat, aes(x = x, y = p)) +
  geom_linerange(aes(ymin = 0, ymax = p), linewidth = 3) +
  theme_bw() +
  geom_hline(yintercept = 0) +
  ylab('density') +
  labs(subtitle = 'n = 15, p = 0.50') +
  xlim(0,15)
```

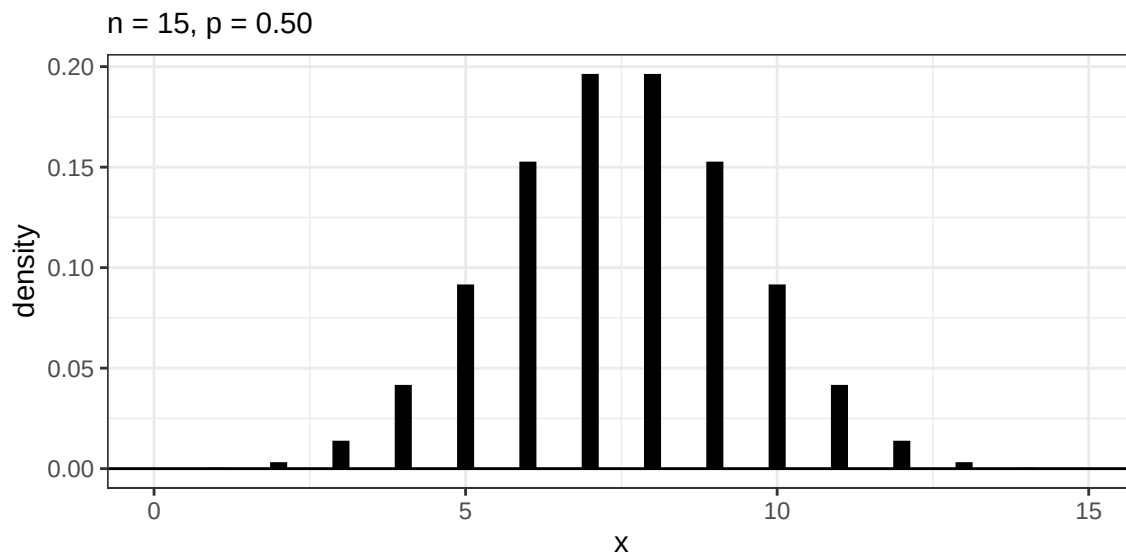


Figure 1: Binomial Distribution Plot

- ii. What is the $\alpha = 0.05$ percentile from this distribution?

```
alpha = 0.05 #intended alpha
qbinom(p = alpha, size = n, prob = p) #inverse cdf to find 5th percentile observation
## [1] 4
```

The $\alpha = 0.05$ percentile from this distribution is 4 according to *qbinom*.

$$b_\alpha = F_B^{-1}(\alpha)$$

iii. Compute $P(B \leq b_\alpha)$. Is it in the rejection region?

```
pbinom(q = 4, size = n, prob = p)
## [1] 0.05923462
# find the associated probability of observing 4 or fewer from our null distribution
```

It is not in the rejection region because it is not less than 0.05.

iv. What is the largest observation in the rejection region?

```
pbinom(q = 3, size = n, prob = p) # Test out 3
## [1] 0.01757813
```

3 is the largest observation in the rejection region.

v. What is the achieved α for this test? That is, what is the probability of a Type I error for this specification?

The achieved α is 0.018, because that is essentially our threshold for significance when the largest rejectable value has that probability associated.

(b) A z-test.

i. Plot the standard Gaussian distribution.

```
# create tibble of gaussian densities
gaussian_dat <- tibble(x = seq(from = -5,
                              to = 5,
                              length.out = 10000),
                      d = dnorm(seq(from = -5,
                                    to = 5,
                                    length.out = 10000)))

ggplot(data = gaussian_dat, aes(x = x, y = d)) + geom_line() +
  theme_bw() +
  xlab('Z') +
  ylab('density') +
  geom_hline(yintercept = 0)
```

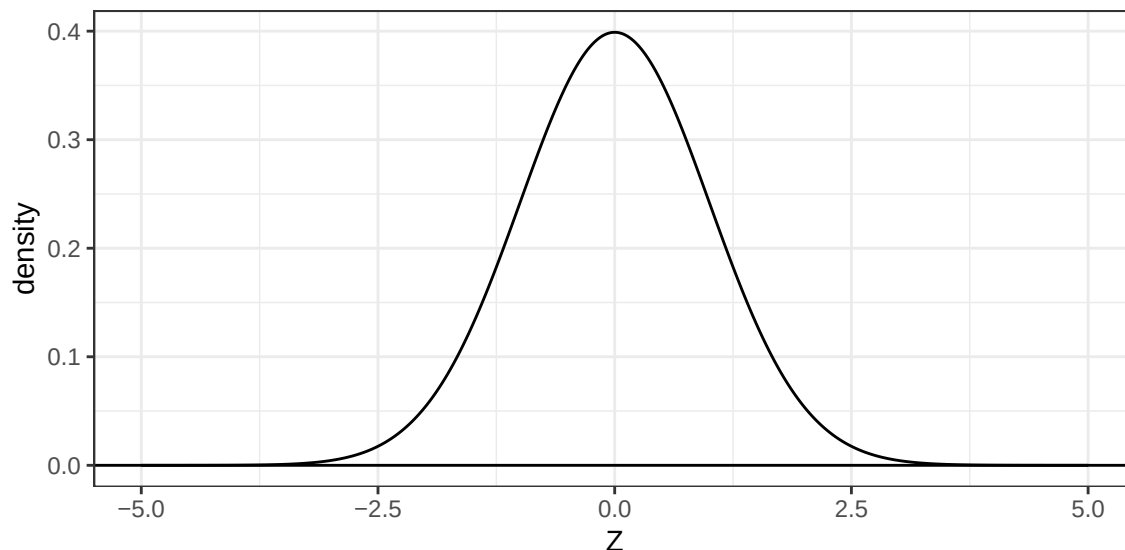


Figure 2: Gaussian Distribution Plot

ii. What is the $\alpha = 0.05$ percentile from this distribution?

$$z_\alpha = F_Z^{-1}(\alpha)$$

```
fifth_percentile <- qnorm(p = alpha)

fifth_percentile
## [1] -1.644854
```

The 5th percentile observation is -1.645 standard deviations from the mean.

- iii. Compute $P(Z \leq z_\alpha)$. Is it in the rejection region?

```
pnorm(q = fifth_percentile)
## [1] 0.05
```

Technically this observation is not in the rejection region, as we would consider anything with a probability of < 0.05 .

- iv. What is the largest observation in the rejection region?

Hint: We know z , n and p_0 , solve for $\hat{p}$.

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

$$-1.644854 > \frac{\hat{p} - 0.5}{\sqrt{0.5(1 - 0.5)/15}}$$

$$-1.644854 > \frac{\hat{p} - 0.5}{0.129099}$$

$$-0.212350 > \hat{p} - 0.5$$

$$0.287650 > \hat{p}$$

$$\frac{4}{15} = 0.2\bar{6}$$

A proportion < 0.287650 would be in the rejection region. The largest observation given $n = 15$ would be 4.

- v. What is the achieved α for this test? That is, what is the probability of a Type I error for this specification?

```
p.hat = 4/15
p0 = 0.5
n = 15

(z.stat <- (p.hat - p0)/sqrt(p0*(1-p0)/n))
## [1] -1.807392
pnorm(q = z.stat)
## [1] 0.03535057
```

The achieved α for this test is approximately 0.035.

2. Suppose we have data that follow an exponential distribution, i.e.,

$$X_1, X_2, \dots, X_n \sim \text{Exponential}(\lambda).$$

When n is large, we can employ the Central Limit Theorem to assert the following approximation

$$T = \frac{\bar{X} - \lambda}{s/\sqrt{n}} \sim \mathcal{AT}(v = n - 1).$$

Recall that this approximation is “good” when n is large and the underlying population distribution is more symmetric. The skewness of the Exponential distribution is 2, regardless of λ , so the approximation really leans on the sample size in this case.

Conveniently, when the population distribution is Exponential, mathematical statisticians can derive the *exact* sampling distribution of $\bar{X}$,

$$\bar{X} \sim \text{Gamma} \left(\alpha = n, \beta = \frac{n}{\lambda} \right).$$

If you don't want to take my word for it, see the small simulation below where $n = 10$ and $\lambda = 2$.

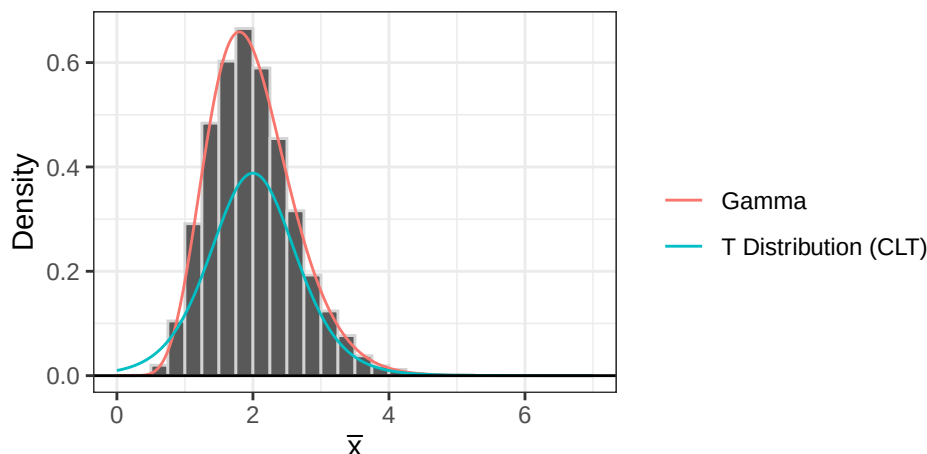


Figure 3: A simulation showing that even for small n the exact Gamma sampling distribution fits well.

How can I use the Gamma sampling distribution to derive an exact test for $\mu_X = E(X) = \lambda$ when we assume the population distribution is Exponential?

- (a) What should the hypotheses be? Assume a two-sided test.

$$H_0 : \bar{X} = \lambda = \lambda_0$$

$$H_a : \bar{X} = \lambda \neq \lambda_0$$

- (b) What should the test statistic be?

The test statistic would be $\bar{X}$.

- (c) What is the sampling distribution of the statistic under the null hypothesis?

The sampling distribution of $\bar{X}$ under H_0 is a Gamma distribution with $\alpha = n$ and $\beta = \frac{n}{\lambda_0}$.

- (d) What values of $\bar{X}$ provide statistically discernible support for the alternative hypothesis?

Values of $\bar{X}$ that provide statistically discernible support for H_a would be those with which the cumulative probability under the null distribution of observing such an $\bar{X}$ or more extreme is $< \frac{\alpha}{2}$ (for a two sided test).

- (e) How would you compute the p -value?

For a two sided test, an approximate method would be to double the p -value you got from a function like `pgamma()`. There's no way to exactly count it because the Gamma distribution is non-symmetric, so for a more applicable method I'd use randomization testing or bootstrapping.

3. Compute the probability we make an observation in the rejection region when $\lambda = 2$ and $n = 10$ under the following conditions.

- (a) When $\lambda = 2$, what is the probability we make an observation below the 2.5th percentile or above the 97.5th percentile (as computed in Question 2)?

```

n = 10
lambda = 2

low_percentile = 0.025
high_percentile = 0.975

(low_percentile_value <- qgamma(p = low_percentile, shape = n, rate = n/lambda))

## [1] 0.9590777

(p_low_percentile <- mean(xbar < low_percentile_value))

## [1] 0.0253

(high_percentile_value <- qgamma(p = high_percentile, shape = n, rate = n/lambda))

## [1] 3.416961

(p_high_percentile <- mean(xbar > high_percentile_value))

## [1] 0.0239

(p_low_percentile + p_high_percentile)

## [1] 0.0492

```

P(making an observation below the 2.5th percentile or above the 97.5th percentile | $\lambda = 2$) = 0.0492.

- (b) When $\lambda = 2.3$, what is the probability we make an observation in between the 2.5th percentile and the 97.5th percentile (as computed in Question 2)?

```

n = 10
lambda = 2.3

low_percentile = 0.025
high_percentile = 0.975

(low_percentile_value <- qgamma(p = low_percentile, shape = n, rate = n/lambda))

## [1] 1.102939

(high_percentile_value <- qgamma(p = high_percentile, shape = n, rate = n/lambda))

## [1] 3.929505

(p_between <- mean((xbar > low_percentile_value) & (xbar < high_percentile_value)))

## [1] 0.9384

```

P(making observation between 2.5th percentile and 97.5th percentile | $\lambda = 2.3$) = 0.9384.

- (c) When $\lambda = 2.3$, what is the probability we make an observation below the 2.5th percentile or above the 97.5th percentile (as computed in Question 2)?

```

(p_outside = 1 - p_between)

## [1] 0.0616

```

P(making observation below the 2.5th percentile or above the 97.5th percentile | $\lambda = 2.3$) = 0.0616.

- (d) Redo (a)-(c), except now consider the case where we have misspecified the population distribution. Suppose that the true underlying population distribution is

$$X_i \sim \text{Gaussian}(\mu_X = \lambda, \sigma_X = \lambda),$$

and therefore

$$\bar{X} \sim \text{Gaussian}(\mu_{\bar{X}} = \lambda, \sigma_{\bar{X}} = \lambda/\sqrt{n}).$$

```

# a again

n = 10
lambda = 2

low_percentile = 0.025
high_percentile = 0.975

(low_percentile_value <- qnorm(p = low_percentile, mean = lambda, sd = lambda/sqrt(n)))

## [1] 0.7604099

(p_low_percentile <- mean(xbar < low_percentile_value))

## [1] 0.0058

(high_percentile_value <- qnorm(p = high_percentile, mean = lambda, sd = lambda/sqrt(n)))

## [1] 3.23959

(p_high_percentile <- mean(xbar > high_percentile_value))

## [1] 0.04

(p_low_percentile + p_high_percentile)

## [1] 0.0458

# b again

lambda = 2.3

(low_percentile_value <- qnorm(p = low_percentile, mean = lambda, sd = lambda/sqrt(n)))

## [1] 0.8744714

(high_percentile_value <- qnorm(p = high_percentile, mean = lambda, sd = lambda/sqrt(n)))

## [1] 3.725529

(p_between <- mean((xbar > low_percentile_value) & (xbar < high_percentile_value)))

## [1] 0.9749

# c again

(p_outside = 1 - p_between)

## [1] 0.0251

```

Given that $\bar{X}$ is actually normally distributed:

- (a) the probability of making an observation outside of the 2.5th and 97.5th percentile when $\lambda = 2$ is 0.0458.
 - (b) the probability of making an observation between the 2.5th percentile and 97.5th percentile given $\lambda = 2.3$ is 0.9749.
 - (c) the probability of making an observation outside of the 2.5th percentile and 97.5th percentile given $\lambda = 2.3$ is 0.0251.
4. Complete the following to assess the coverage of different methods for constructing a confidence interval for p .

In a loop, generate an observation from the Binomial($n = 15$, $p = 0.10$) distribution and compute the Pearson Clopper interval and the Wilson interval ten thousand times.

- (a) What proportion of the time is the true proportion in the Pearson Clopper interval? The Wilson interval?

```
library(binom)
set.seed(1313)

correct_pearson <- rep(NA, 10000)
correct_wilson <- rep(NA, 10000)

for(i in 1:10000){

  random_binom <- rbinom(n = 15, size = 1, prob = 0.10)

  successes <- sum(random_binom)

  conf_ints <- binom.confint(x = successes, n = 15, methods = c("exact", "wilson"))

  pearson_ci <- c(conf_ints$lower[1], conf_ints$upper[1])

  wilson_ci <- c(conf_ints$lower[2], conf_ints$upper[2])

  correct_pearson[i] <- ifelse(0.1 >= pearson_ci[1] && 0.1 <= pearson_ci[2], 1, 0)

  correct_wilson[i] <- ifelse(0.1 >= wilson_ci[1] && 0.1 <= wilson_ci[2], 1, 0)

}

mean(correct_pearson)
## [1] 0.9871

mean(correct_wilson)
## [1] 0.9455
```

The proportion of Pearson Clopper intervals that have the true proportion is 0.9871. The Wilson intervals have a proportion of 0.9455.

- (b) Consider $n = 1000$ and $p = 0.10$. What proportion of the time is the true proportion in the Pearson Clopper interval? The Wilson interval?

```
correct_pearson <- rep(NA, 10000)
correct_wilson <- rep(NA, 10000)
```

```

for(i in 1:10000){
  random_binom <- rbinom(n = 1000, size = 1, prob = 0.10)
  successes <- sum(random_binom)
  conf_ints <- binom.confint(x = successes, n = 1000, methods = c("exact", "wilson"))
  pearson_ci <- c(conf_ints$lower[1], conf_ints$upper[1])
  wilson_ci <- c(conf_ints$lower[2], conf_ints$upper[2])
  correct_pearson[i] <- ifelse(0.1 >= pearson_ci[1] && 0.1 <= pearson_ci[2], 1, 0)
  correct_wilson[i] <- ifelse(0.1 >= wilson_ci[1] && 0.1 <= wilson_ci[2], 1, 0)
}
mean(correct_pearson)
## [1] 0.959
mean(correct_wilson)
## [1] 0.9528

```

The proportion of Pearson Clopper intervals that capture the true proportion is 0.959, while the Wilson intervals is 0.9528.

- (c) Consider $n = 15$ and $p = 0.50$. What proportion of the time is the true proportion in the Pearson Clopper interval? The Wilson interval?

```

correct_pearson <- rep(NA, 10000)
correct_wilson <- rep(NA, 10000)

for(i in 1:10000){
  random_binom <- rbinom(n = 15, size = 1, prob = 0.50)
  successes <- sum(random_binom)
  conf_ints <- binom.confint(x = successes, n = 15, methods = c("exact", "wilson"))
  pearson_ci <- c(conf_ints$lower[1], conf_ints$upper[1])
  wilson_ci <- c(conf_ints$lower[2], conf_ints$upper[2])
  correct_pearson[i] <- ifelse(0.5 >= pearson_ci[1] && 0.5 <= pearson_ci[2], 1, 0)
  correct_wilson[i] <- ifelse(0.5 >= wilson_ci[1] && 0.5 <= wilson_ci[2], 1, 0)
}
mean(correct_pearson)
## [1] 0.9643

```



```
mean(correct_wilson)
## [1] 0.9643
```

The proportion of Pearson Clopper intervals that capture the true proportion is 0.964, while the Wilson intervals is 0.9643.

- (d) Consider $n = 5$ and $p = 0.50$. What proportion of the time is the true proportion in the Pearson Clopper interval? The Wilson interval?

```
correct_pearson <- rep(NA, 10000)
correct_wilson <- rep(NA, 10000)

for(i in 1:10000){

  random_binom <- rbinom(n = 5, size = 1, prob = 0.50)

  successes <- sum(random_binom)

  conf_ints <- binom.confint(x = successes, n = 5, methods = c("exact", "wilson"))

  pearson_ci <- c(conf_ints$lower[1], conf_ints$upper[1])

  wilson_ci <- c(conf_ints$lower[2], conf_ints$upper[2])

  correct_pearson[i] <- ifelse(0.5 >= pearson_ci[1] && 0.5 <= pearson_ci[2], 1, 0)

  correct_wilson[i] <- ifelse(0.5 >= wilson_ci[1] && 0.5 <= wilson_ci[2], 1, 0)

}

mean(correct_pearson)
## [1] 1

mean(correct_wilson)
## [1] 0.9362
```

The proportion of Pearson Clopper intervals that capture the true proportion is 1, while the Wilson intervals is 0.9362.

- (e) What do parts (a)-(d) tell us about when to use each approach?

From testing the types of intervals at varying sample sizes and proportions, we find that the Wilson interval is more consistently achieving the desired alpha and capturing the true population proportion. We find that the Pearson Clopper interval is often too wide and doesn't achieve the desired alpha, especially at smaller sample sizes.

5. **(Optional)** During office hours, I explained a statistically discernible result as “an indicator that the alternative hypothesis is true or an unusual event.” Describe what this means and why it is accurate.
6. **(Optional)** Continuing with Questions 2-3. In some instances, it is easier to approximate the probability of Type I and Type II errors, and Power using simulation.
 - (a) Conduct a simulation study that evaluates the proportion of Type I errors and Power by filling in the table below. Note this assumes the *true* value of λ is 2.
Hint: Note that you're repeating the same task for different n and λ . Using a function or a loop should help tremendously here.

Sample Size	True λ	Proportion Rejected (Exact)	Proportion Rejected (CLT)
5	2 (type I)		
	2.1		
	2.2		
	2.3		
	2.4		
10	2 (type I)		
	2.1		
	2.2		
	2.3		
	2.4		
20	2 (type I)		
	2.1		
	2.2		
	2.3		
	2.4		
50	2 (type I)		
	2.1		
	2.2		
	2.3		
	2.4		

- (b) Redo this simulation. Except, now, generate data from a Gaussian distribution instead of an Exponential distribution. Use $\mu_X = \sigma_X = \lambda$. What happens to the results?

7. **(Optional)** Continuing with Questions 2-3. Pouyiourou et al. (2023) explores cancer of unknown primary (CUP). As part of their work, they measure the concentration of cell-free DNA (ccfDNA), which is strongly linked to cancer prognosis. They hypothesize that ccfDNA will be higher in CUP patients, which would indicate a worse prognosis.

The data can be loaded as follows for $n = 34$ CUP patients.

```
dat <- c(4.14, 1.23, 3.28, 4.58, 1.33, 1.93, 7.97, 7.99, 15.30, 14.00, 4.31, 14.87,
        28.08, 11.18, 12.56, 2.00, 1.26, 149.02, 0.27, 1.45, 3.13, 0.76, 15.02,
        5.36, 32.60, 22.08, 261.17, 6.61, 2.43, 5.02, 1.16, 5.94, 3.36, 15.91)
```

- (a) Plot the data and compute corresponding statistics to begin to assess the claim that the population mean is greater than 2.
- (b) Conduct a t -test to see whether there is support for the claim that the population mean is different than 2 among this group. Note that 2 is the estimated average among healthy individuals. Ensure to check all conditions for use.
- (c) Conduct your exact test to see whether there is support for the claim that the population mean is different than 2 among this group. Note that 2 is the estimated average among healthy individuals. Ensure to check all conditions for use.
- (d) Your solution to parts (b) and (c) should differ. Why do they differ, and which result do you trust most?
- Hint:** If you are having trouble, consider performing a bootstrap or randomization test.
- (e) Recognizing that the data are rather skewed, instead conduct a test about the population median. Is the population median greater than 1.75?

References

Pouyiourou, M., Kraft, B. N., Wohlfromm, T., Stahl, M., Kubuschok, B., Löffler, H., Hacker, U. T., Hübner, G., Weiss, L., Bitzer, M., et al. (2023). Nivolumab and Ipilimumab in recurrent or refractory cancer of unknown primary: A phase II trial. *Nature Communications*, 14(1):6761.